

X


<https://swayam.gov.in>

https://swayam.gov.in/nc_details/NPTEL

riya.soni17oct@gmail.com ▾

NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction To Machine Learning - IITKGP

(course)



Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

☐ Practice: Week 0 : Assignment 0 (assessment? name=141)

☒ Solution of Assignment 0 (unit? unit=5&lesson=115)

Week 1 ()

Week 2 ()

Week 3 ()

Week 0 : Assignment 0

Assignment not submitted

1)

2 points

Find the maxima and minima of the function $f(x) = x + \frac{1}{x}$.

- A. -1,1
 B. 1,-1
 C. -2,2
 D. 2,-2

- ☐ A.
☐ B.
☐ C.
☐ D.

2)

2 points

Precision is defined as the fraction of relevant instances among the retrieved instances and Recall is defined as the fraction of relevant instances that have been retrieved over the total amount of relevant instances. A typical Information Retrieval system retrieves a total of 20 documents for a particular query out of which only 5 are relevant. Find the Precision and Recall of the system. Total set of relevant documents = 10.

- A. 0.5,0.25
 B. 0.25, 0.5
 C. 0.5,0.5
 D. 0.25,0.25

- ☐ A.
☐ B.
☐ C.
☐ D.

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Assignment
Solution ()Download
Videos ()Problem
Solving
Session -
July 2024 ()

3)

2 points

Entropy associated with each possible data value is the negative logarithm of the probability mass function for the value. Example Formula is:

$$H(S) = - \sum_i p_i \log_2(p_i)$$

Here, $H(S)$ denotes entropy, i represents a class, and p_i denotes the probability of that class.

Given a list of 20 examples including 10 positive, 5 negative and 5 neutral examples. The entropy of the dataset with respect to this classification is:

- A. 3/2
- B. 2
- C. 5/2
- D. 3

- ☐ A.
- ☐ B.
- ☐ C.
- ☐ D.

4)

2 points

Find the limit $\lim_{x \rightarrow 2} \frac{\sqrt{7+x}-3}{x-2}$
(Hint: Use L-Hospital's rule)

- A. 1/3
- B. 1/6
- C. 2/3
- D. 5/6

- ☐ A.
- ☐ B.
- ☐ C.
- ☐ D.

5)

2 points

5 runners run a race. How many different ways can the top 3 finishers be selected, if we do not care about the specific order of these top 3?

- A. 5
- B. 10
- C. 20
- D. 30

- ☐ A.
- ☐ B.
- ☐ C.
- ☐ D.

6)

2 points

A busy student must complete 3 problem sets before doing laundry. Each problem set requires 1 day with probability $2/3$ and 2 days with probability $1/3$. Let B be the number of days a busy student delays laundry. What is $E[B]$?

[Here, $E[B]$ denotes the expectation of the event B]

Example: If the first problem set requires 1 day and the second and third problem sets each requires 2 days, then the student delays for $B = 5$ days.

- A. 2
- B. 3
- C. 4
- D. 5

- ☐ A.
- ☐ B.
- ☐ C.
- ☐ D.

7)

2 points

In a class, there are 15 students who like chocolate. 13 students like vanilla. 10 students like neither. If there are 35 students in the class, how many students like chocolate and vanilla?

- A. 2
- B. 12
- C. 3
- D. 20

- ☐ A.
- ☐ B.
- ☐ C.
- ☐ D.

8)

2 points

Suppose there is a sentence "let's play or not play". The bag-of-words representation vector of the sentence is the count of each word in the sentence, which corresponds to:

let's play or not
[1 2 1 1].

The point in the space is $s = (1, 2, 1, 1)$.

Now suppose we have some query vectors related to 'play' $q_1 = (0, 1, 0, 0)$ and a query vector related to 'let's' $q_2 = (1, 0, 0, 0)$. Find the nearest query of the sentence vector (s).
Hint : (Use Cosine similarity distance of two points to perform the same).

- A. q_1
- B. q_2

- ☐ A.
- ☐ B.

9)

2 points

Let u be a $n \times 1$ vector, such that $u^T u = 1$. Let I be the $n \times n$ identity matrix. The $n \times n$ matrix A is given by $(I - k u u^T)$, where k is a real constant. u itself is an eigenvector of A , with eigenvalue -1 . What is the value of k ?

A. -2

B. -1

C. 2

D. 0

☐ A.☐ B.☐ C.☐ D.

10)

2 points

Let $A^{m \times n}$ be a matrix of real numbers. The matrix AA^T has an eigenvector x with eigenvalue b . Then the eigenvector y of $A^T A$ which has eigenvalue b is equal to

A. $x^T A$ B. $A^T x$ C. x D. Cannot be described in terms of x ☐ A.☐ B.☐ C.☐ D.**Check Answers and Submit**